

Tutorial 6/Module 2

Transient Analysis of First and Second Order Circuit

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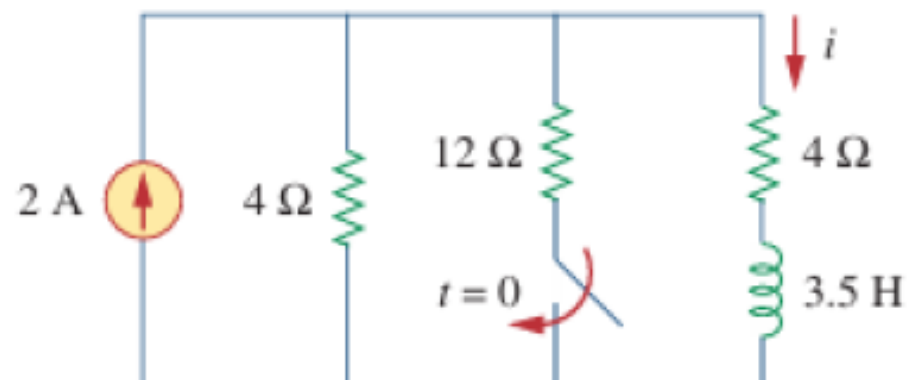
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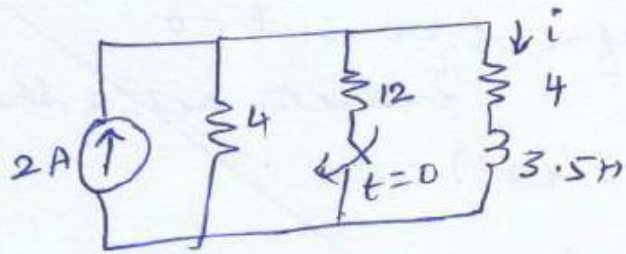
Tutorial 1

Determine the inductor current for both $t < 0$ and $t > 0$ in the circuits of Fig.



Answer Key

(3)



$$R_{eq} = 4 + (4 \parallel 12) = 7\Omega$$

$$i(0) = 1$$

$$i(\infty) = \frac{2 \times (4 \parallel 12)}{4 + (4 \parallel 12)} = \frac{6}{7}$$

$$i(t) = \frac{6}{7} + \left[1 - \frac{6}{7} \right] e^{-t/0.5}$$

$$\therefore \underline{i(t) = \frac{1}{7} (6 - e^{-2t}) A.}$$

$t < 0$

$$i(t) = \frac{2 \times 4}{4 + 4} = 1 A.$$

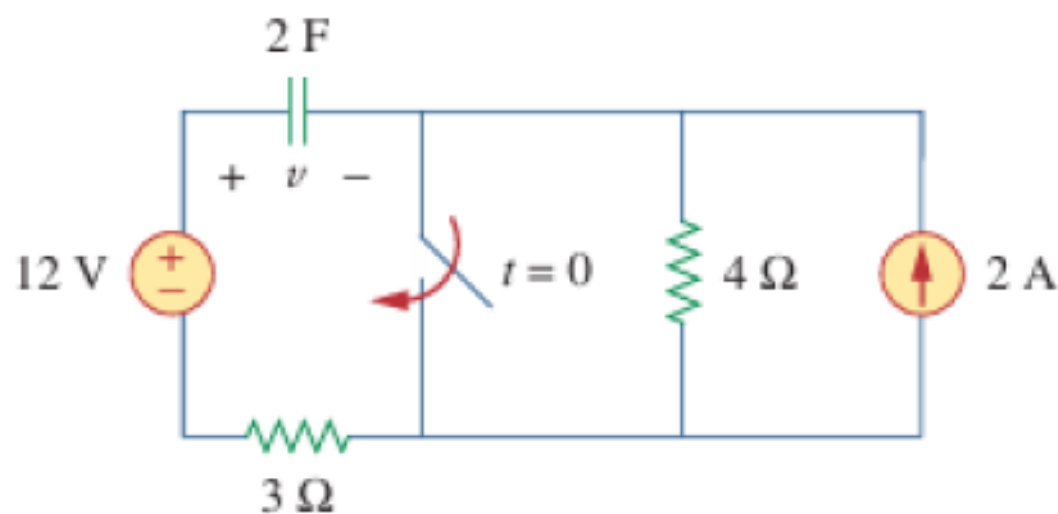
$t > 0$

$$i(t) = i(\infty) + [i(0) - i(\infty)] e^{-t/\tau}$$

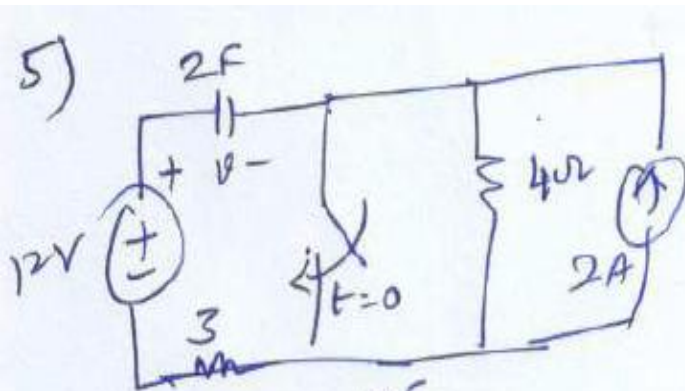
$$\tau = \frac{L}{R_{eq}} = \frac{3.5}{7} = \frac{1}{2}$$

Tutorial 2

Calculate the capacitor voltage for $t < 0$ and $t > 0$ in the circuit of Fig.



Answer Key

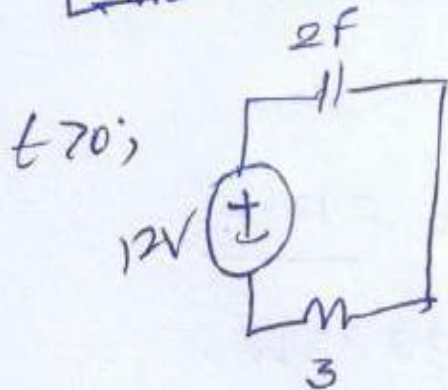
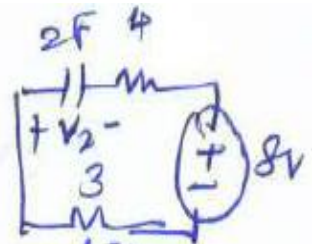


$$t < 0; \quad V = V_1 + V_2 \text{ (SPT)}$$

Due to 12V Source : $V_1 = 12V$

Due to 2A Source — do Source Transformation

$$V_2 = -8V.$$



~~$$V(t) = V(\infty) + [V(0) - V(\infty)]e^{-t/\tau}$$~~

$$V = 12 - 8 = 4V.$$

$$V(t) = V(\infty) + [V(0) - V(\infty)]e^{-t/\tau}$$

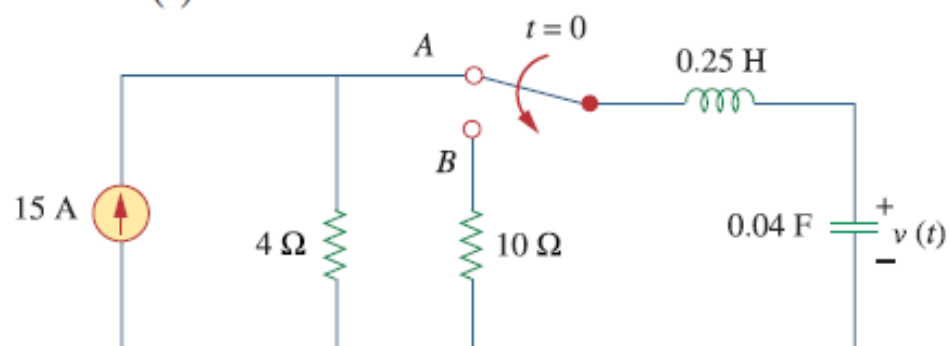
$$V(\infty) = 12 \quad V(0) = 4. \quad \tau = RC = 6$$

$$V(t) = 12 + (4 - 12)e^{-t/6}$$

$$V(t) = 12 - 8e^{-t/6} \text{ V}$$

Tutorial 3

In the circuit of Fig. 8.71, the switch instantaneously moves from position A to B at $t = 0$. Find $v(t)$ for all $t \geq 0$.



Answer Key

$$i(0) = I_0 = 0, \quad v(0) = V_0 = 4 \times 15 = 60$$

$$\frac{di(0)}{dt} = -\frac{1}{L}(RI_0 + V_0) = -4(0 + 60) = -240$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{\frac{1}{4} \frac{1}{25}}} = 10$$

$$\alpha = \frac{R}{2L} = \frac{10}{2 \frac{1}{4}} = 20, \text{ which is } > \omega_0.$$

$$s = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -20 \pm \sqrt{300} = -20 \pm 10\sqrt{3} = -2.679, -37.32$$

$$i(t) = A_1 e^{-2.679t} + A_2 e^{-37.32t}$$

$$i(0) = 0 = A_1 + A_2, \quad \frac{di(0)}{dt} = -2.679A_1 - 37.32A_2 = -240$$

$$\text{This leads to } A_1 = -6.928 = -A_2$$

$$i(t) = 6.928(e^{-37.32t} - e^{-2.679t})$$

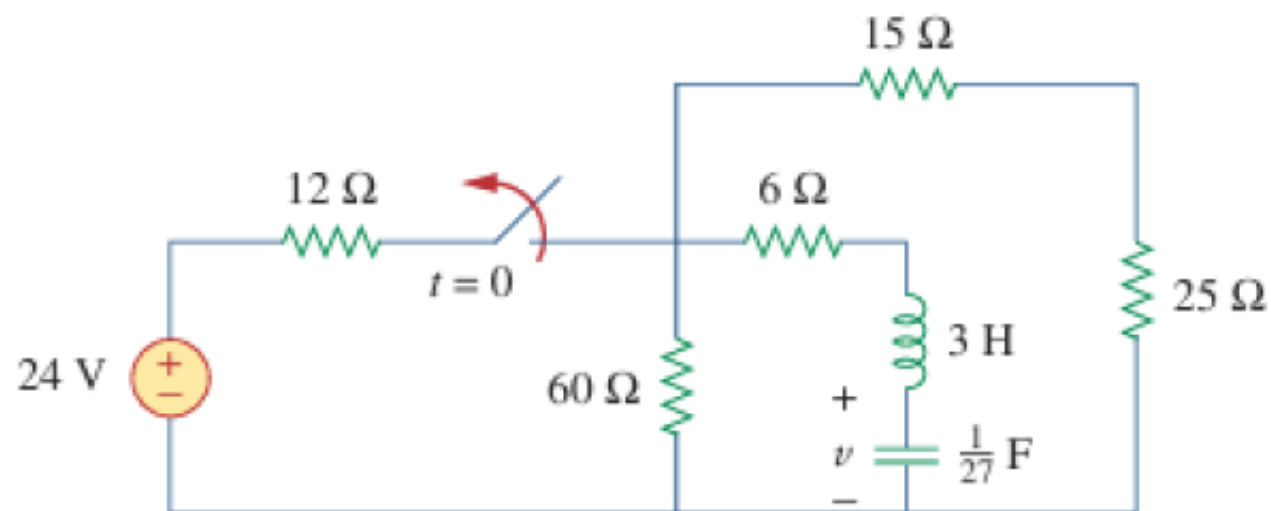
$$\text{Since, } v(t) = \frac{1}{C} \int_0^t i(t) dt + \text{const, and } v(0) = 60V, \text{ we get}$$

$$v(t) = \underline{(64.65e^{-2.679t} - 4.641e^{-37.32t}) \text{ V}}$$

We note that $v(0) = 60.009V$ and not $60V$. This is due to rounding errors since $v(t)$ must go to zero as time goes to infinity. {In other words, the constant of integration must be zero.

Tutorial 4

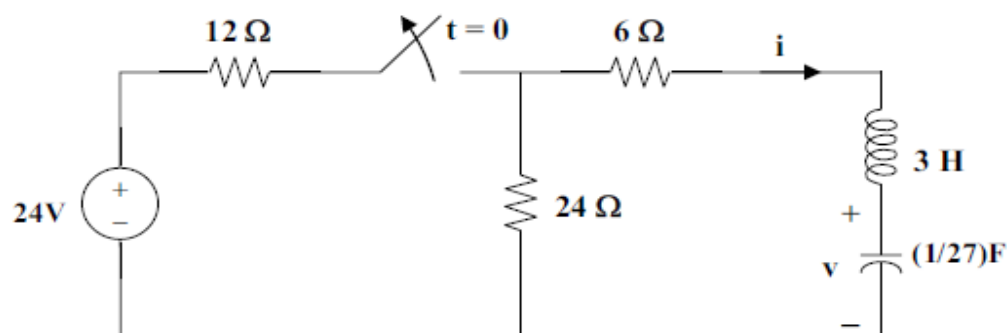
5. Calculate $v(t)$ for $t > 0$ in the circuit .



Answer Key

By combining some resistors, the circuit is equivalent to that shown below.

$$60 \parallel (15 + 25) = 24 \text{ ohms.}$$



$$\text{At } t = 0^-, \quad i(0) = 0, \quad v(0) = 24 \times 24 / 36 = 16 \text{ V}$$

For $t > 0$, we have a series RLC circuit. $R = 30 \text{ ohms}$, $L = 3 \text{ H}$, $C = (1/27) \text{ F}$

$$\alpha = R/(2L) = 30/6 = 5$$

$$\omega_o = 1/\sqrt{LC} = 1/\sqrt{3 \times 1/27} = 3, \text{ clearly } \alpha > \omega_o \text{ (overdamped response)}$$

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_o^2} = -5 \pm \sqrt{5^2 - 3^2} = -9, -1$$

$$v(t) = [Ae^{-t} + Be^{-9t}], \quad v(0) = 16 = A + B \quad (1)$$

$$i = Cdv/dt = C[-Ae^{-t} - 9Be^{-9t}]$$

$$i(0) = 0 = C[-A - 9B] \text{ or } A = -9B \quad (2)$$

From (1) and (2), $B = -2$ and $A = 18$.

$$\text{Hence,} \quad v(t) = \underline{(18e^{-t} - 2e^{-9t}) \text{ V}}$$

Thank You